Introduction - Homework 3

- 2D Backwards Heat Equation
  - Application on Image
  - Self-adjointness
  - Landweber Iterations
- Total Variation Inpainting
  - TV Inpainting
  - Harmonic Inpainting
  - Noise vs Noiseless
- Feedback received on report
  - Mis-stated heat equation as convolution
    - Not strictly a convolution since kernel is not shift-invariant
    - Speed-up is similar to convolution, we turn sum-of-sums into pointwise multiplication

2D Backwards Heat Equation

We consider the 2D backwards heat equation on domain $\Omega = (0, \pi) \times (0, \pi)$ with homogeneous Dirichlet boundary conditions as given by forward operator:

$$Kx(t_1, t_2) = \int_0^\pi \int_0^\pi x(s_1, s_2) k(t_1, t_2, s_1, s_2) \, \mathrm{d}s_1 \, \mathrm{d}s_2$$

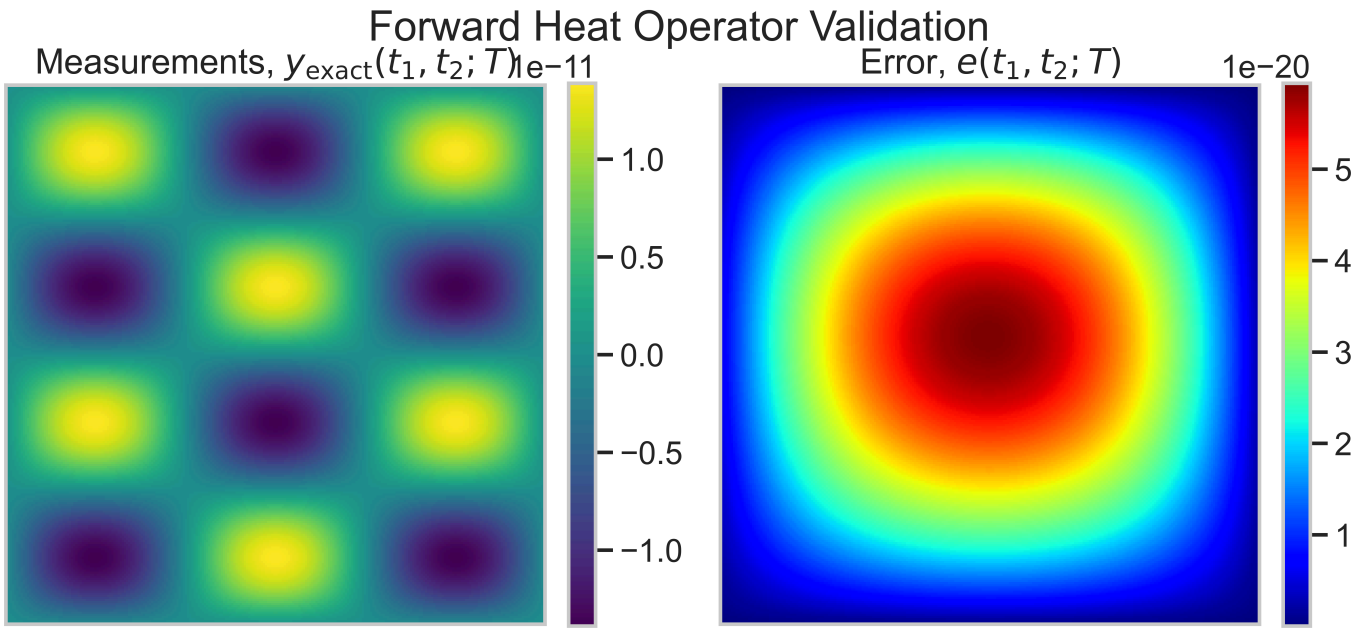
$$k(t_1, t_2, s_1, s_2) = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} e^{-(n^2+m^2)T} \sin(nt_1) \sin(mt_2) \sin(ns_1) \sin(ms_2)$$

- T is terminal time
- Consider Gaussian decay in kernel:
  - Higher $T \Longrightarrow$ more ill-posed
  - Higher modes $n, m \Longrightarrow$ more ill-posed
- Problem is satisfies uniqueness and existence in Hadamard sense, but **not** stability

2D Backwards Heat Equation — Implementation

- Homogenous Dirichlet BCs work well with sine basis
- We model only the interior of domain since boundary is fixed
- Can be done as matrix-free operator with 2D FFT ($n \log n$ complexity)
  - This requires odd-extension to produce sine basis: “Unfold” input by mirroring in both dimensions
  - Apply 2D FFT, multiply in frequency domain, apply inverse 2D FFT
  - Alternative is to use Discrete Sine Transform (DST), could prevent odd-extension and lower memory allocations
  - Math simple: $y = \mathcal{F}^{-1}\left(e^{-(n^2+m^2)T} \cdot \mathcal{F}(x)\right)$
  - Implementation a bit tricker due to wave-numbers, orderings, odd-extension

2D Backwards Heat Equation — Validation of Implementation



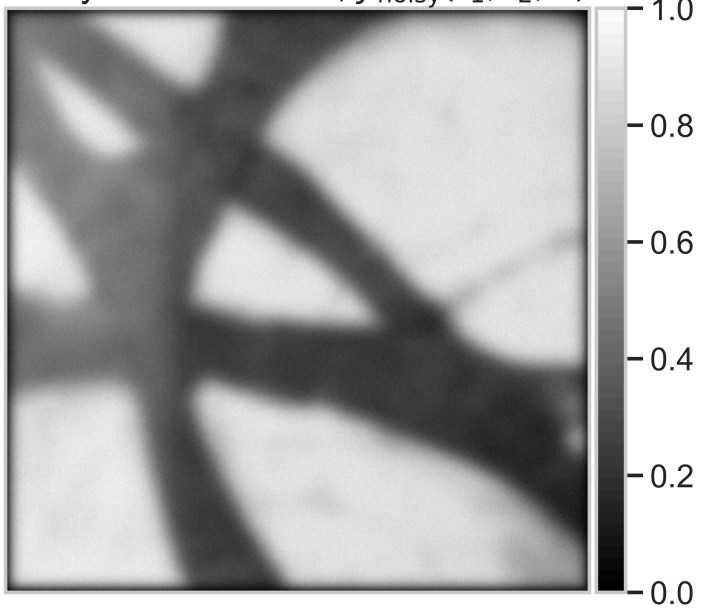
- Test using single mode
- $x_{\text{test}}(s_1, s_2) = \sin(ks_1)\sin(ls_2)$
- $k = 4, l = 3$
- Forward operator just attenuates by $e^{-(k^2+l^2)T}$
- Observe excellent agreement
- Error $e \sim 10^{-20}$

2D Backwards Heat Equation — Apply to Image

Initial condition, $x(s_1, s_2)$



Noisy measurements, $y_{\text{noisy}}(t_1, t_2; T)$



- Apply to an image
- Forward model just blurs image according to diffusion
- Terminal time $T = 0.001$
- Noise level $\sigma = 10^{-2}$

2D Backwards Heat Equation — Self-Adjointness

Self-adjointness of operator K is easily proven from symmetry of kernel

$$\begin{aligned} Kx(t_1,t_2) &= \int_0^\pi \int_0^\pi k(t_1,t_2,s_1,s_2)x(s_1,s_2)\,\mathrm{d}s_1\,\mathrm{d}s_2 \\ &= \int_0^\pi \int_0^\pi k^*(\bar{s}_1,\bar{s}_2,t_1,t_2)x(s_1,s_2)\,\mathrm{d}s_1\,\mathrm{d}s_2 = K^*x(t_1,t_2), \end{aligned}$$

where k^* is adjoint of the kernel

$$\begin{aligned} k(t_1,t_2,s_1,s_2) &= \sum_{n,m} \beta_{nm} \sin(ns_1)\sin(ms_2)\sin(nt_1)\sin(mt_2) \\ &= k(s_1,s_2,t_1,t_2) = k^*(\bar{s}_1,\bar{s}_2,t_1,t_2) \end{aligned}$$

with $\bar{s}_i = s_i$ since $s_i \in \mathbb{R}$ and $\beta_{nm} = e^{-(n^2+m^2)T}$

Thus, $K = K^*$

2D Backwards Heat Equation — Landweber Iterations

Landweber iterations are also known as *gradient descent* and are given by the iteration

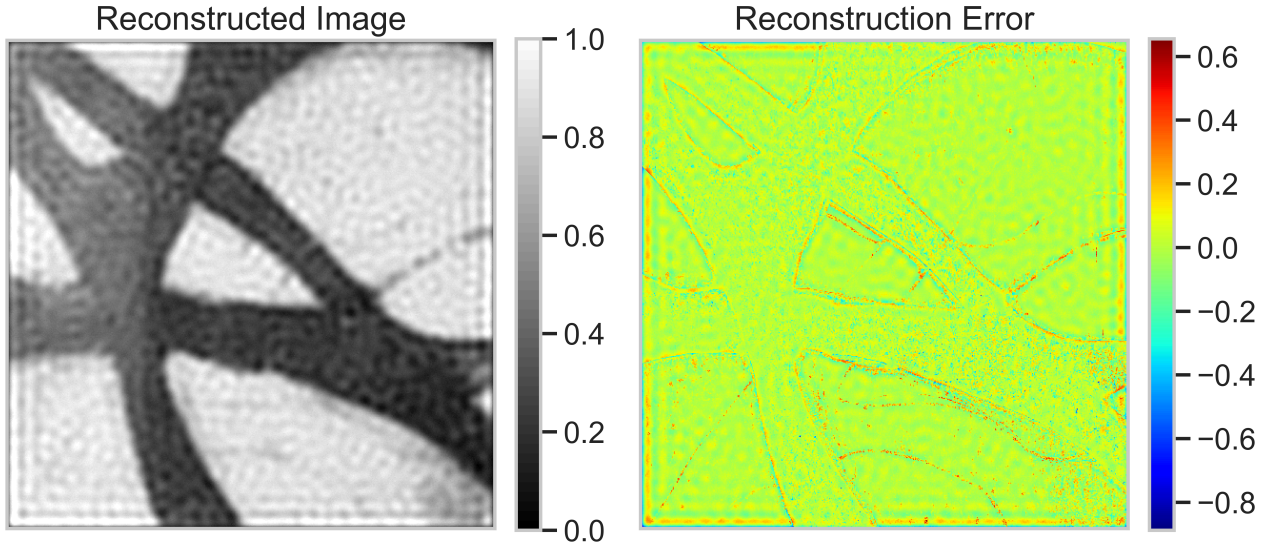
$$x_{j+1} = x_j + \omega K^* \left(K x_j - y^\delta \right)$$

- Discretisation and self-adjointness of K yields $x_{j+1} = x_j + \omega A \left(A x_j - y^\delta \right)$
- Relaxation parameter ω must satisfy $0 < \omega < 2/\sigma_{\max}^2$
  - We may find this analytically as $\sigma_{\max} = e^{-2T}$
  - We let $\omega = \frac{18}{10} e^{4T}$ to have safety margin of $\frac{9}{10}$
- Method is *semi-convergent*
  - Finds local minimum
  - Early stopping regularises solution and prevents overfitting to noise

With this, we are able to reconstruct image from blurred and noisy observation.

2D Backwards Heat Equation — Landweber Reconstruction

- Note wave-like artefacts near boundary due to image not satisfying homogenous Dirichlet BCs
- These are reminiscent of the familiar Gibbs phenomenon
- Error concentrated on high-frequency edges as expected
- Does not converge to exact solution
- Tried CGLS, quickly diverges (≈3 iterations)



Inpainting

We recall the two inpainting models considered:

$$\min_u \frac{1}{2}\left\|\mathbb{1}_{\Omega\setminus\mathcal{M}}(z-u)\right\|^2 + \lambda \int_\Omega |\nabla u(x)|_2 \,\mathrm{d}x \qquad \text{TV Inpainting}$$

$$\min_{u\in H_0^1(\Omega)} \frac{1}{2}\left\|\mathbb{1}_{\Omega\setminus\mathcal{M}}(z-u)\right\|^2 + \frac{\lambda}{2}\|\nabla u(x)\|^2 \,\mathrm{d}x \qquad \text{Harmonic Inpainting}$$

Where λ is a regularisation parameter and z is the the remaining parts of the corrupted image as given by

$$z = \begin{cases} \bar{u} + \eta & \text{in } \Omega \setminus \mathcal{M} \\ \text{unknown} & \text{in } \mathcal{M} \end{cases}$$

Note that in the noiseless case $\eta = 0$ the data-fidelity term vanishes if imposing known pixels as hard constraints, which allows the regularisation parameter λ to be absorbed.

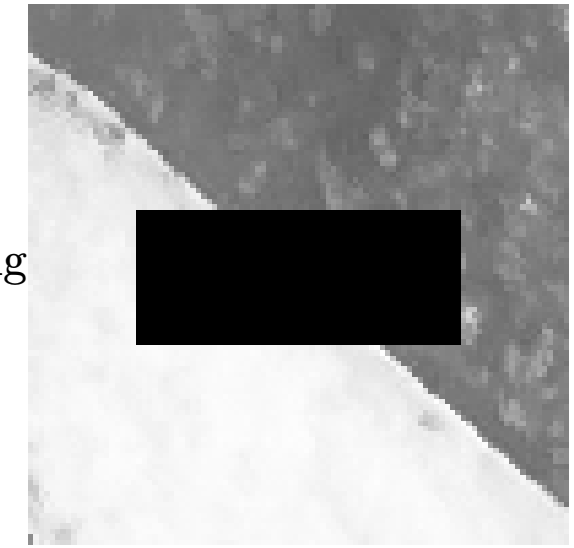
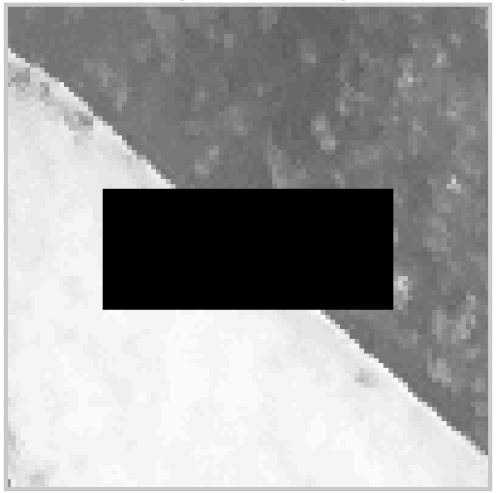


Figure 1: Image to be inpainted

TV Inpainting

The TV inpainting is carried out using `cvxpy` in both the regular case and the case of assumed noiselessness.

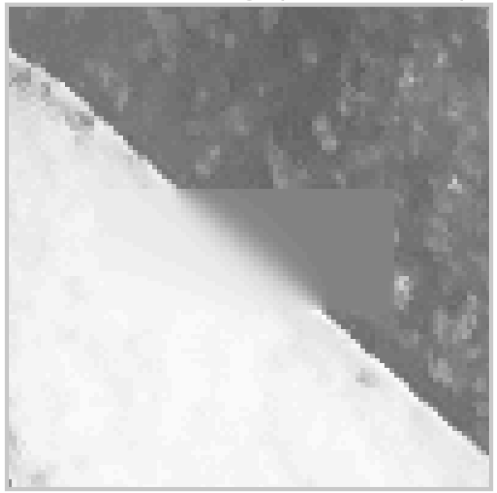
Original Image



TV Inpainting ($\lambda = 0.01$)



TV Inpainting (Noiseless)

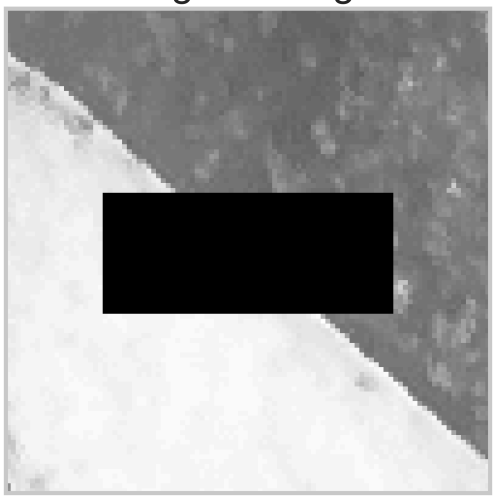


- TV tries to minimise total variation, i.e., the length of edges
- TV liable to staircasing. $\partial\mathcal{M}$ is one *"stair"*
- Good candidate for TV due to sharp, straight edge
- Noiseless case seems preferable here

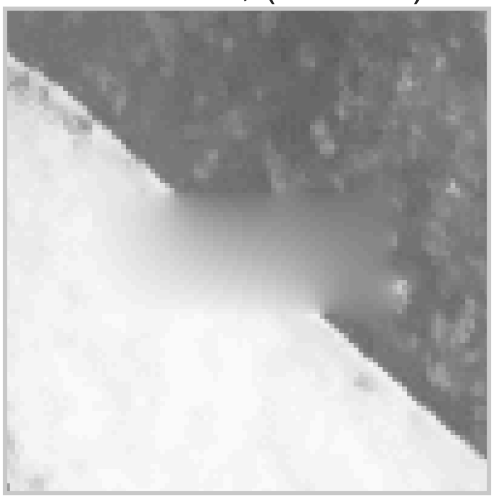
Harmonic Inpainting

Harmonic inpainting is carried out in a similar fashion using `cvxpy`, again for both cases.

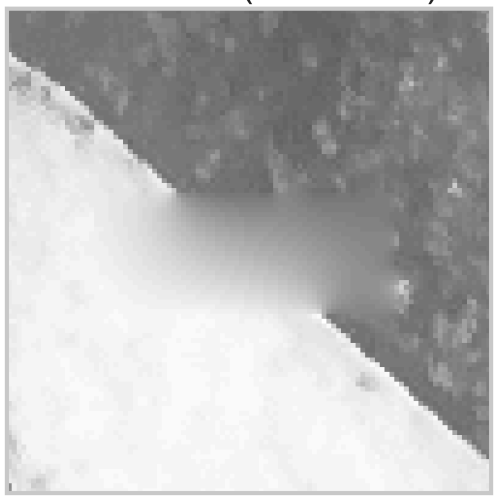
Original Image



Harmonic, ($\lambda = 0.1$)



Harmonic (Noiseless)



- Harmonic inpainting minimises gradients $\Longrightarrow$ smooth solutions
- This is better at smoothing transition along $\partial\mathcal{M}$
- Significantly worse recovery of sharp, straight edge
- Noiseless case again seems preferable